ORF 524 — Midterm Exam

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Contents

Notes:

- Duration of examination: 180 minutes.
  An additional 15 minutes are given to upload your solutions to Gradescope.

- All parts are equally weighted (5 points).
  Please provide as much detail as possible in your answers.
  Answers without proper justification will not receive full credit.

- Please start each question on a separate page.
  A single PDF file must be submitted to Gradescope.

- Only one page of personal notes are allowed during the examination.
  This is otherwise a closed-book and individual examination.

1 Uniform Most Powerful Testing (50 points)

Assume $\mathbf{X} = (X_1, \ldots, X_n)^\top \sim \mathbb{P}_\theta$ with $\theta \in \Theta$. Let $L(\mathbf{x}|\theta)$ denote the likelihood function of $\mathbb{P}_\theta$. Recall that a *critical region* (or *rejection region*) $\mathcal{C}$ is a subset of the sample space such that the null hypothesis H_0 is rejected if (and only if) the observed sample $\mathbf{X}$ falls into $\mathcal{C}$.

The celebrated **Neyman-Pearson Lemma** states that for testing $\mathsf{H}_0 : \theta = \theta_0$ vs. $\mathsf{H}_1 : \theta = \theta_1$ at a significance level $\alpha \in (0, 1)$, the most powerful critical region $\mathcal{C}$ is defined by

$$\mathcal{C}_{\texttt{NP}} = \left\{ \mathbf{x} : \frac{L(\mathbf{x}|\theta_1)}{L(\mathbf{x}|\theta_0)} > k \right\},$$

where k is chosen such that $\mathbb{P}_{\theta_0}(\mathbf{X} \in \mathcal{C}_{\texttt{NP}}) = \alpha$. (For the purpose of this question, we focus on the core non-randomized rejection region $\mathcal{C}_{\texttt{NP}}$ and ignore the boundary case where the ratio equals k, which occurs with $\mathbb{P}_{\theta_0}$-zero probability when $\mathbf{X}$ is continuously distributed.)

The following four questions will guide in proving the **Neyman-Pearson Lemma**. Let $\mathcal{C}$ be any other critical region such that $\mathbb{P}_{\theta_0}(\mathbf{X} \in \mathcal{C}) = \alpha$.

1. Define the power of a test associated with the critical region $\mathcal{C}$, denoted by $\beta_{\mathcal{C}}(\theta_1)$.

2. Decompose the difference between the rejection regions $\mathcal{C}_{\texttt{NP}}$ and $\mathcal{C}$ into the disjoint regions $\mathcal{C}_{\texttt{NP}} \setminus \mathcal{C}$ and $\mathcal{C} \setminus \mathcal{C}_{\texttt{NP}}$. Use the size constraint to show that: $\beta_{\mathcal{C}_{\texttt{NP}} \setminus \mathcal{C}}(\theta_0) = \beta_{\mathcal{C} \setminus \mathcal{C}_{\texttt{NP}}}(\theta_0)$.

3. Prove that the integral of $L(\mathbf{x}|\theta_1)$ over the region $\mathcal{C}_{\texttt{NP}} \setminus \mathcal{C}$ is greater to the integral of $L(\mathbf{x}|\theta_1)$ over the region $\mathcal{C} \setminus \mathcal{C}_{\texttt{NP}}$. (It would be "greater or equal" if accounting for the boundary case via randomization was needed.)

4. Prove the **Neyman-Pearson Lemma** by showing that $\beta_{\mathcal{C}_{\texttt{NP}}}(\theta_1) > \beta_{\mathcal{C}}(\theta_1)$.

For the rest of this question, assume that $\mathbf{X} = (X_1, \ldots, X_n)^\top$ is a vector of i.i.d. random variables with marginal distribution $\mathsf{Exponential}(\theta)$, and $\theta_1 > \theta_0$.

5. Write down the joint likelihood function $L(\mathbf{x}|\theta)$ and the likelihood ratio $\Lambda(\mathbf{x}) = \frac{L(\mathbf{x}|\theta_1)}{L(\mathbf{x}|\theta_0)}$.

6. Find the most powerful testing procedure for $\mathsf{H}_0 : \theta = \theta_0$ vs. $\mathsf{H}_1 : \theta = \theta_1$.

7. Note that $T = \sum_{i=1}^n X_i \sim \mathsf{Gamma}(n, \theta_0)$. For a test of size α, derive the critical value c in terms of n, θ_0, and an appropriate quantile of the Gamma distribution.

8. Consider the specific scenario where $n = 1$, $\theta_0 = 1$, and $\alpha = 0.05$. Find the exact numerical value of the critical constant c. Then, calculate the power of this test when the true parameter is $\theta_1 = 2$.

9. Find the uniform most powerful (UMP) testing procedure for $\mathsf{H}_0 : \theta \leq \theta_0$ vs. $\mathsf{H}_1 : \theta > \theta_1$.

10. Why is the critical value c derived derived for the simple null $\mathsf{H}_0 : \theta = \theta_0$ still the correct critical value to use for the composite null $\mathsf{H}_0 : \theta \leq \theta_0$ to guarantee that the UMP test has size α?

2 Misspecified Maximum Likelihood (50 points)

Let $\{X_i : 1, 2, \cdots, n\}$ be a random sample from $X \sim \mathbb{P}_{\theta_0}$ with $\mathbb{P}_{\theta_0} \in \mathcal{P} = \{\mathbb{P}_\theta : \theta \in \Theta \subseteq \mathbb{R}\}$, where $F(x) = \mathbb{P}_\theta[X_i \leq x]$ is absolutely continuous with Lebesgue density $f(x;\theta)$. Let $f(x) = f(x;\theta_0)$ to save notation. Consider another family of Lebesgue densities, $\{g(x;\theta) : \theta \in \Theta \subseteq \mathbb{R}\}$, which may not be equal to $f(x;\theta)$. A *possibly misspecified* Maximum Likelihood Estimator (MLE) is

$$\breve{\theta}_n = \arg\max_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^{n} \log g(X_i;\theta),$$

and the associated population optimization problem is

$$\breve{\theta}_0 = \arg\max_{\theta \in \Theta} \mathbb{E}_{\theta_0}[\log g(X;\theta)] = \arg\max_{\theta \in \Theta} \int [\log g(x;\theta)] f(x) dx.$$

assuming the "usual" regularity conditions hold. You will be asked to state and impose additional assumptions (or regularity conditions) as needed below.

1. Providing appropriate regularity conditions, show that

$$\mathbb{E}_{\theta_0}\left[\log\left(\frac{g(X;\theta)}{f(X)}\right)\right] = \int \log\left(\frac{g(x;\theta)}{f(x)}\right) f(x) dx \leq 0,$$

   with equality if only if $g(x;\theta) = f(x)$.

2. Use this result to justify the following interpretation for the pseudo-true value $\breve{\theta}_0$:

$$\breve{\theta}_0 = \arg\min_{\theta \in \Theta} D(g(\cdot,\theta), f), \qquad D(g(\cdot,\theta), f) = \int [\log f(x)] f(x) dx - \int [\log g(x;\theta)] f(x) dx.$$

   The "metric" $D(g(\cdot,\theta), f)$ is called the *Kullbach-Leibler* (K-L) distance. Can you give an interpretation to the (possibly) misspecified MLE based on this criterion?

3. Providing appropriate regularity conditions, show that

$$\mathbb{E}_{\theta_0}\left[\frac{\partial}{\partial\theta} \log g(X;\theta)\bigg|_{\theta=\breve{\theta}_0}\right] = 0.$$

   What happens when $g(x;\breve{\theta}_0) = f(x)$ or when $g(x;\theta) = f(x;\theta)$?

4. Providing appropriate regularity conditions, show that

$$\sqrt{n}(\breve{\theta}_n - \breve{\theta}_0) = -\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \psi(X_i) + o_{\mathbb{P}}(1) \to_d \mathcal{N}(0, V(\breve{\theta}_0)),$$

   where $\mathbb{E}_{\theta_0}[\psi(X_i)] = 0$ and $\mathbb{E}_{\theta_0}[\psi(X_i)^2] = V(\breve{\theta}_0)$.
   Give the exact form of the influence function $\psi(x)$.

5. Show that $V(\breve{\theta}_0) = H(\breve{\theta}_0)^{-1}\Sigma(\breve{\theta}_0)H(\breve{\theta}_0)^{-1}$, and give the exact form of the variance components $H(\theta)$ and $\Sigma(\theta)$.
   What happens when $g(x;\breve{\theta}_0) = f(x)$ or when $g(x;\theta) = f(x;\theta)$?

6. Propose a consistent estimator of the asymptotic variance $V(\breve{\theta}_0)$.
   Construct a feasible, asymptotically pivotal statistic.
   What happens when $g(x;\breve{\theta}_0) = f(x)$ or when $g(x;\theta) = f(x;\theta)$?

7. Suppose that $X_i = \mathbb{1}(\theta_0 - \epsilon_i \geq 0)$ with $\{\epsilon_i : 1, 2, \cdots, n\}$ be a random sample from $\epsilon \sim \mathcal{N}(0,1)$.

   (a) Show that $\mathcal{P} = \{\mathsf{Bernoulli}(\Phi(\theta)) : \theta \in \Theta \subseteq \mathbb{R}\}$.
   Find the (correctly specified) maximum likelihood estimator $\hat{\theta}_{\texttt{ML}}$ of θ_0.
   Find the asymptotic distribution of $\sqrt{n}(\hat{\theta}_{\texttt{ML}} - \theta_0)$, and give the exact form of the asymptotic variance component(s).

   (b) Suppose a researcher employs the misspecified model $\epsilon \sim \mathsf{Uniform}(0,1)$.
   Find the misspecified maximum likelihood estimator $\breve{\theta}_n$ of $\breve{\theta}_0$, and compute the form of pseudo-true value $\breve{\theta}_0$ in this case.
   Find the asymptotic distribution of $\sqrt{n}(\breve{\theta}_n - \breve{\theta}_0)$, and give the exact form of the asymptotic variance component(s).

8. Consider the hypothesis testing problem:

$$\mathsf{H}_0 : g(x;\breve{\theta}_0) = f(x) \qquad \text{vs.} \qquad \mathsf{H}_1 : g(x;\breve{\theta}_0) \neq f(x).$$

   In words, H_0 corresponds to correct specification of the likelihood function, and H_1 corresponds to misspecification of the likelihood function.

   (a) Under minimal regularity conditions,

$$-\mathbb{E}_{\theta_0}\left[\frac{\partial^2}{\partial\theta^2}\log g(X;\theta)\bigg|_{\theta=\breve{\theta}_0}\right] \neq \mathbb{E}_{\theta_0}\left[\left(\frac{\partial}{\partial\theta}\log g(X;\theta)\bigg|_{\theta=\breve{\theta}_0}\right)^2\right]$$

   unless $g(x;\breve{\theta}_0) = f(x)$. Using this result, propose an infeasible hypothesis test for misspecification of the likelihood function.

   (b) Propose a feasible testing procedure for misspecification of the likelihood function.

Table 1.1. Discrete Distributions on $\mathcal{R}$

Uniform	p.d.f.	$1/m$, $x = a_1, ..., a_m$
	m.g.f.	$\sum_{j=1}^m e^{a_j t}/m$, $t \in \mathcal{R}$
$DU(a_1, ..., a_m)$	Expectation	$\sum_{j=1}^m a_j/m$
	Variance	$\sum_{j=1}^m (a_j - \bar{a})^2/m$, $\bar{a} = \sum_{j=1}^m a_j/m$
	Parameter	$a_i \in \mathcal{R}$, $m = 1, 2, ...$
Binomial	p.d.f.	$\binom{n}{x} p^x (1-p)^{n-x}$, $x = 0, 1, ..., n$
	m.g.f.	$(pe^t + 1 - p)^n$, $t \in \mathcal{R}$
$Bi(p, n)$	Expectation	np
	Variance	$np(1-p)$
	Parameter	$p \in [0, 1]$, $n = 1, 2, ...$
Poisson	p.d.f.	$\theta^x e^{-\theta}/x!$, $x = 0, 1, 2, ...$
	m.g.f.	$e^{\theta(e^t - 1)}$, $t \in \mathcal{R}$
$P(\theta)$	Expectation	θ
	Variance	θ
	Parameter	$\theta > 0$
Geometric	p.d.f.	$(1-p)^{x-1}p$, $x = 1, 2, ...$
	m.g.f.	$pe^t/[1 - (1-p)e^t]$, $t < -\log(1-p)$
$G(p)$	Expectation	$1/p$
	Variance	$(1-p)/p^2$
	Parameter	$p \in [0, 1]$
Hyper-	p.d.f.	$\binom{n}{x}\binom{m}{r-x} / \binom{N}{r}$
geometric		$x = 0, 1, ..., \min\{r, n\}$, $r - x \le m$
	m.g.f.	No explicit form
$HG(r, n, m)$	Expectation	rn/N
	Variance	$rnm(N-r)/[N^2(N-1)]$
	Parameter	$r, n, m = 1, 2, ...$, $N = n + m$
Negative	p.d.f.	$\binom{x-1}{r-1} p^r (1-p)^{x-r}$, $x = r, r+1, ...$
binomial	m.g.f.	$p^r e^{rt}/[1 - (1-p)e^t]^r$, $t < -\log(1-p)$
	Expectation	r/p
$NB(p, r)$	Variance	$r(1-p)/p^2$
	Parameter	$p \in [0, 1]$, $r = 1, 2, ...$
Log-	p.d.f.	$-(\log p)^{-1} x^{-1} (1-p)^x$, $x = 1, 2, ...$
distribution	m.g.f.	$\log[1 - (1-p)e^t]/\log p$, $t \in \mathcal{R}$
	Expectation	$-(1-p)/(p \log p)$
$L(p)$	Variance	$-(1-p)[1 + (1-p)/\log p]/(p^2 \log p)$
	Parameter	$p \in (0, 1)$

All p.d.f.'s are w.r.t. counting measure.

Table 1.2. Distributions on $\mathcal{R}$ with Lebesgue p.d.f.'s

Uniform	p.d.f.	$(b-a)^{-1}I_{(a,b)}(x)$
	m.g.f.	$(e^{bt}-e^{at})/[(b-a)t],\ \ t\in\mathcal{R}$
$U(a,b)$	Expectation	$(a+b)/2$
	Variance	$(b-a)^2/12$
	Parameter	$a,\ b\in\mathcal{R},\ \ a<b$
Normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}e^{-(x-\mu)^2/2\sigma^2}$
	m.g.f.	$e^{\mu t+\sigma^2t^2/2},\ \ t\in\mathcal{R}$
$N(\mu,\sigma^2)$	Expectation	μ
	Variance	σ^2
	Parameter	$\mu\in\mathcal{R},\ \ \sigma>0$
Exponential	p.d.f.	$\theta^{-1}e^{-(x-a)/\theta}I_{(a,\infty)}(x)$
	m.g.f.	$e^{at}(1-\theta t)^{-1},\ \ t<\theta^{-1}$
$E(a,\theta)$	Expectation	$\theta+a$
	Variance	θ^2
	Parameter	$\theta>0,\ \ a\in\mathcal{R}$
Chi-square	p.d.f.	$\frac{1}{\Gamma(k/2)2^{k/2}}x^{k/2-1}e^{-x/2}I_{(0,\infty)}(x)$
	m.g.f.	$(1-2t)^{-k/2},\ \ t<1/2$
χ^2_k	Expectation	k
	Variance	$2k$
	Parameter	$k=1,2,...$
Gamma	p.d.f.	$\frac{1}{\Gamma(\alpha)\gamma^\alpha}x^{\alpha-1}e^{-x/\gamma}I_{(0,\infty)}(x)$
	m.g.f.	$(1-\gamma t)^{-\alpha},\ \ t<\gamma^{-1}$
$\Gamma(\alpha,\gamma)$	Expectation	$\alpha\gamma$
	Variance	$\alpha\gamma^2$
	Parameter	$\gamma>0,\ \alpha>0$
Beta	p.d.f.	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}I_{(0,1)}(x)$
	m.g.f.	No explicit form
$B(\alpha,\beta)$	Expectation	$\alpha/(\alpha+\beta)$
	Variance	$\alpha\beta/[(\alpha+\beta+1)(\alpha+\beta)^2]$
	Parameter	$\alpha>0,\ \ \beta>0$
Cauchy	p.d.f.	$\frac{1}{\pi\sigma}\left[1+\left(\frac{x-\mu}{\sigma}\right)^2\right]^{-1}$
	ch.f.	$e^{\sqrt{-1}\mu t-\sigma\|t\|}$
$C(\mu,\sigma)$	Expectation	Does not exist
	Variance	Does not exist
	Parameter	$\mu\in\mathcal{R},\ \ \sigma>0$

Table 1.2. (continued)

t-distribution	p.d.f.	$\frac{\Gamma[(n+1)/2]}{\sqrt{n\pi}\Gamma(n/2)}\left(1+\frac{x^2}{n}\right)^{-(n+1)/2}$
	ch.f.	No explicit form
t_n	Expectation	$0,\ \ (n>1)$
	Variance	$n/(n-2),\ \ (n>2)$
	Parameter	$n=1,2,...$
F-distribution	p.d.f.	$\frac{n^{n/2}m^{m/2}\Gamma[(n+m)/2]x^{n/2-1}}{\Gamma(n/2)\Gamma(m/2)(m+nx)^{(n+m)/2}}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$F_{n,m}$	Expectation	$m/(m-2),\ \ (m>2)$
	Variance	$2m^2(n+m-2)/[n(m-2)^2(m-4)]$, $(m>4)$
	Parameter	$n=1,2,...,\ \ m=1,2,...$
Log-normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}x^{-1}e^{-(\log x-\mu)^2/2\sigma^2}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$LN(\mu,\sigma^2)$	Expectation	$e^{\mu+\sigma^2/2}$
	Variance	$e^{2\mu+\sigma^2}(e^{\sigma^2}-1)$
	Parameter	$\mu\in\mathcal{R},\ \ \sigma>0$
Weibull	p.d.f.	$\frac{\alpha}{\theta}x^{\alpha-1}e^{-x^\alpha/\theta}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$W(\alpha,\theta)$	Expectation	$\theta^{1/\alpha}\Gamma(\alpha^{-1}+1)$
	Variance	$\theta^{2/\alpha}\left\{\Gamma(2\alpha^{-1}+1)-\left[\Gamma(\alpha^{-1}+1)\right]^2\right\}$
	Parameter	$\theta>0,\ \ \alpha>0$
Double	p.d.f.	$\frac{1}{2\theta}e^{-\lvert x-\mu\rvert/\theta}$
Exponential	m.g.f.	$e^{\mu t}/(1-\theta^2t^2),\ \ \lvert t\rvert<\theta^{-1}$
	Expectation	μ
$DE(\mu,\theta)$	Variance	$2\theta^2$
	Parameter	$\mu\in\mathcal{R},\ \ \theta>0$
Pareto	p.d.f.	$\theta a^\theta x^{-(\theta+1)}I_{(a,\infty)}(x)$
	ch.f.	No explicit form
$Pa(a,\theta)$	Expectation	$\theta a/(\theta-1),\ \ (\theta>1)$
	Variance	$\theta a^2/[(\theta-1)^2(\theta-2)],\ \ (\theta>2)$
	Parameter	$\theta>0,\ \ a>0$
Logistic	p.d.f.	$\sigma^{-1}e^{-(x-\mu)/\sigma}/[1+e^{-(x-\mu)/\sigma}]^2$
	m.g.f.	$e^{\mu t}\Gamma(1+\sigma t)\Gamma(1-\sigma t),\ \lvert t\rvert<\sigma^{-1}$
$LG(\mu,\sigma)$	Expectation	μ
	Variance	$\sigma^2\pi^2/3$
	Parameter	$\mu\in\mathcal{R},\ \sigma>0$